UNIT 1

INTRODUCTION TO SEMICONDUCTOR PHYSICS AND DEVICE TECHNOLOGY

Band Theory

When individual atoms come together to form a solid, their atomic energy levels no longer remain isolated. In a single atom, electrons occupy discrete energy levels such as $1s$, $2s$, or $3p$. However, in a solid crystal where atoms are closely packed in a periodic arrangement, the proximity of neighboring atoms causes these discrete levels to split into a very large number of closely spaced states. This happens because of quantum mechanical interactions, particularly the Pauli exclusion principle, which prevents electrons from occupying the same quantum state. The result is the formation of continuous ranges of energy known as **energy bands**.

The lower energy orbitals (such as s orbitals) combine to form the **valence band**, which is usually filled with electrons at low temperatures. The higher energy orbitals (such as p orbitals) form the **conduction band**, which can host electrons that are free to move and contribute to electrical conduction. Between these two bands there may exist an **energy gap** or **band gap**, a forbidden region where no electron states exist. The size of this band gap is the key factor that determines whether a material behaves as a conductor, semiconductor, or insulator. In conductors (metals), the conduction and valence bands overlap, or the valence band itself is only partially filled, which allows electrons to move freely. In semiconductors and insulators, the valence band and conduction band are separated by a finite band gap; this gap is small in semiconductors (allowing limited conduction under certain conditions) and large in insulators (effectively preventing conduction).

Thus, band theory provides a quantum mechanical explanation for why some materials conduct electricity easily, others conduct only under special conditions, and some do not conduct at all.

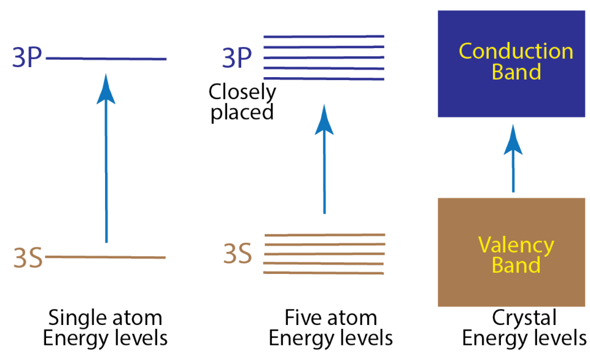

Figure 1.1: Formation of bands from discrete atomic levels.

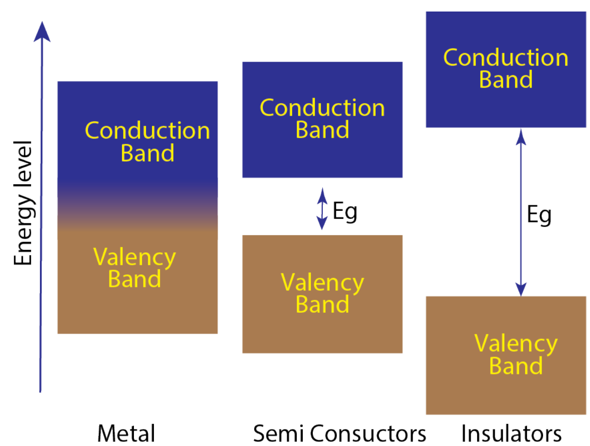

Figure 1.2: Energy band structures for metals, semiconductors, and insulators.

1.1 Conductors (Metals)

In metals, the electronic band structure allows electrons to move with great freedom. The essential feature of conductors is that there is **no significant band gap**. In some metals, the valence band is only partially filled, leaving vacant energy states very close to the filled ones. Even a small electric field can push electrons into these nearby empty states, enabling them to move and conduct current. In other metals, the valence band is completely filled but overlaps with the conduction band, meaning electrons can move seamlessly between them. In both cases, there is an abundance of mobile electrons, often described as a **sea of electrons** moving within a lattice of positive metal ions.

Because of this, metals exhibit high electrical and thermal conductivity. When an electric field is applied, the free electrons drift, producing a strong current. Their conductivity (σ) is very high, while resistivity (ρ) is correspondingly low. Temperature, however, affects conduction in metals in a negative way: as the temperature increases, the metal ions vibrate more vigorously (these vibrations are called **phonons**), and these vibrations scatter the moving electrons. As a result, the conductivity decreases with rising temperature.

Examples of good conductors include copper, aluminum, gold, silver, and iron. Their applications range from electrical wiring and power transmission lines to electronic component contacts and heat sinks, where their thermal conductivity is equally useful.

1.2 Insulators

Insulators are materials that strongly resist the flow of current. Their key property is a very large band gap, typically greater than $3\,\mathrm{eV}$. In such materials, the valence band is completely filled, while the conduction band is completely empty. Since the thermal energy available at room temperature ($k_B T \approx 0.025\,\mathrm{eV}$) is far too small to excite electrons across such a wide gap, there are essentially no free carriers in the conduction band. This explains their extremely low conductivity and very high resistivity.

Under ordinary conditions, insulators do not allow current to pass. However, if a very strong electric field is applied, electrons can be forcibly pulled from the valence band into the conduction band. This leads to a sudden surge of current called **dielectric breakdown**, which usually damages the material.

Common examples of insulators are diamond (band gap $\approx 5.5\,\mathrm{eV}$), glass, rubber, wood, quartz, and plastics. Their applications are largely in providing safety and controlling current paths: electrical insulation on wires and cables, circuit board substrates, support structures for power lines, and thermal insulation in devices.

1.3 Semiconductors

Semiconductors occupy a middle ground between conductors and insulators. At absolute zero ($0\,\mathrm{K}$), a semiconductor behaves like an insulator, with a filled valence band and empty conduction band. The crucial difference is that the band gap is relatively small, typically between $0.1\,\mathrm{eV}$ and $3\,\mathrm{eV}$. This small gap means that at room temperature, some electrons have enough thermal energy to cross into the conduction band, leaving behind holes in the valence band. Both the electrons in the conduction band and the holes in the valence band act as charge carriers, making semiconductors conductive to a limited but significant degree.

Unlike metals, the conductivity of semiconductors increases with temperature, because more thermal energy excites more electrons across the band gap, increasing the number of carriers. This is why a silicon diode or transistor works better when slightly warm, unlike a copper wire.

1.3.1 Intrinsic Semiconductors

An **intrinsic semiconductor** is a perfectly pure semiconductor crystal with no intentional doping. Its electrical properties are determined solely by the semiconductor material itself, such as silicon (Si) or germanium (Ge). At absolute zero temperature (0 K), an intrinsic semiconductor behaves like an insulator because the valence band is completely filled and the conduction band is completely empty.

At room temperature, however, thermal energy is sufficient to excite some electrons across the relatively small band gap into the conduction band. Each time this happens, an electron is promoted to the conduction band and a corresponding **hole** is left behind in the valence band. Thus, both electrons and holes are generated in equal numbers and act as charge carriers.

The electrical conductivity of an intrinsic semiconductor is therefore due to these thermally generated electron–hole pairs. Because the number of carriers is relatively small, the conductivity of intrinsic semiconductors is limited. The carrier concentration increases strongly with temperature, which is why semiconductors conduct better at higher temperatures.

1.3.2 Extrinsic Semiconductors

An **extrinsic semiconductor** is one whose conductivity has been deliberately enhanced by the addition of very small amounts of impurity atoms, a process called **doping**. Even though the concentration of dopants is typically as low as one part in a million, the effect on conductivity is dramatic. Depending on the type of impurity added, there are two main kinds of extrinsic semiconductors:

- **n-type semiconductor:** When a pentavalent atom such as phosphorus (P) or arsenic (As) is introduced into silicon, four of its five valence electrons form covalent bonds with neighboring silicon atoms, while the fifth electron is loosely bound. This extra electron can be easily excited into the conduction band, thereby increasing the number of negative charge carriers. In n-type materials, electrons are the **majority carriers**, while holes remain the minority carriers.

- **p-type semiconductor:** When a trivalent atom such as boron (B) or gallium (Ga) is added to silicon, it forms covalent bonds using three valence electrons but leaves one bond incomplete, creating a deficiency of an electron, or a **hole**. These holes can move when neighboring electrons shift to fill them, effectively behaving as positive charge carriers. In p-type materials, holes are the **majority carriers**, while electrons are the minority carriers.

The key advantage of extrinsic semiconductors is that their carrier concentration, and hence conductivity, can be precisely controlled by adjusting the doping level. This tunability makes them the essential materials for all semiconductor devices, from diodes and transistors to integrated circuits and photovoltaic cells.

1.3.3 Direct and Indirect Band Gap Semiconductors

Another important classification of semiconductors is based on the nature of their band gap. The band gap not only determines the energy required to excite an electron from the valence band to the conduction band, but also influences how the semiconductor interacts with light.

Direct band gap semiconductors: In a direct band gap material, the minimum of the conduction band and the maximum of the valence band occur at the same momentum (wave vector k) in the band structure. This means that when an electron makes a transition from the conduction band to the valence band, it can directly release its excess energy in the form of a photon. Because no additional momentum is required, radiative recombination (light emission) is highly efficient.

Direct band gap semiconductors are therefore ideally suited for **optoelectronic devices** such as light emitting diodes (LEDs), laser diodes, and photodetectors. Examples include Gallium Arsenide (GaAs), Gallium Nitride (GaN), and Cadmium Sulfide (CdS).

Indirect band gap semiconductors: In an indirect band gap material, the conduction band minimum and the valence band maximum occur at different values of momentum. As a result, when an electron recombines with a hole, the process cannot conserve both energy and momentum by emitting a photon alone. Instead, it requires the involvement of a lattice vibration (phonon) to carry away the extra momentum. This makes radiative recombination much less probable and less efficient.

Indirect band gap semiconductors are therefore poor light emitters, but they are excellent for electronic applications where light emission is not required. The most widely used semiconductor, silicon (Si), as well as germanium (Ge), are indirect band gap materials.

Summary of Direct vs. Indirect Band Gap Semiconductors

- **Direct band gap:** Efficient light emission, used in LEDs, lasers, and optical devices. Examples: GaAs, GaN.
- **Indirect band gap:** Inefficient light emission, used in integrated circuits, transistors, and solar cells. Examples: Si, Ge.

1.4 p-n Junction Diode

The **p-n junction diode** is the most fundamental semiconductor device and is often described as the "building block" of modern electronics. It acts as a one-way valve for current, allowing it to flow easily in one direction but blocking it in the other. A diode is created by joining together two regions of a single semiconductor crystal: one doped to form a **p-type** region, rich in holes (positive charge carriers), and the other doped to form an **n-type** region, rich in electrons (negative charge carriers). The behavior of this simple device can be understood by looking carefully at how the junction forms, how it behaves under different biasing conditions, and how its current–voltage relationship can be described mathematically.

1.4.1 Formation of the p-n Junction

When the p-type and n-type regions are brought into contact, carriers immediately start to move due to the difference in their concentrations. Two important processes occur simultaneously: **diffusion** and **drift**.

Diffusion current arises because there is a high concentration of holes in the p-region and electrons in the n-region. This concentration gradient causes holes to diffuse from the p-side into the n-side, while electrons diffuse from the n-side into the p-side. Whenever an electron meets a hole, they recombine, eliminating both carriers.

As recombination continues, a region near the junction becomes depleted of mobile carriers. This area is known as the **depletion region** or **space charge region**. What remains in this region are immobile, charged ions: positively charged donor ions (N_D^+) on the n-side and negatively charged acceptor ions (N_A^-) on the p-side. The presence of these ions creates a region of fixed charge, even though no free carriers remain.

This separation of charges sets up an internal **electric field**, which points from the positive donor ions on the n-side toward the negative acceptor ions on the p-side. This electric field opposes further diffusion of carriers, giving rise to what is called the **built-in potential barrier** (V_{bi}). At equilibrium, the diffusion current (caused by the carriers' tendency to move from high to low concentration) is exactly balanced by the **drift current** (caused by the electric field sweeping carriers in the opposite direction). For silicon, this built-in potential is typically about 0.7 V, while for germanium it is closer to 0.3 V.

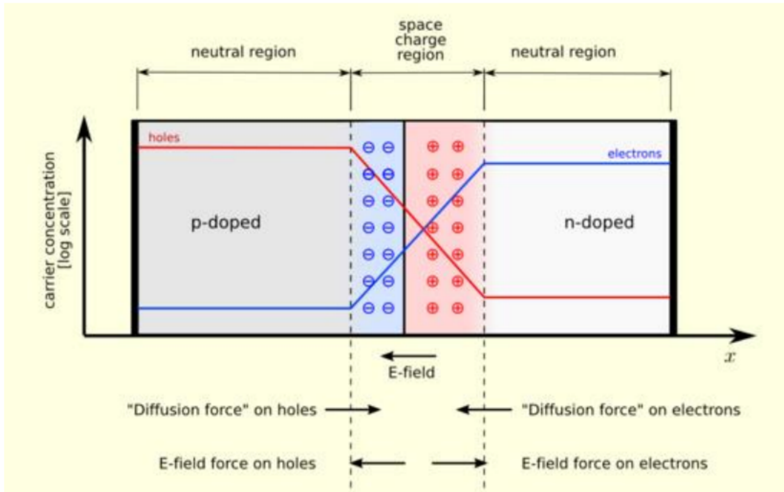

Figure 1.3: Formation of the depletion region at a p-n junction.

1.4.2 Biasing the p-n Junction

The behavior of a diode can be controlled by applying an external voltage, a process called **biasing**. The direction and magnitude of this bias voltage determine whether the diode conducts or blocks current.

In **forward bias**, the positive terminal of the external voltage source is connected to the p-type region and the negative terminal to the n-type region. This applied voltage reduces the effective barrier potential from V_{bi} to $V_{bi} - V_F$. As the voltage increases, the depletion region narrows, and if the applied forward voltage exceeds a certain threshold known as the **cut-in voltage** (approximately 0.7 V for silicon and 0.3 V for germanium), majority carriers are able to cross the junction in large numbers. Electrons from the n-side cross into the p-side, and holes from the p-side cross into the n-side, creating a large forward current (I_F).

In **reverse bias**, the polarity of the source is reversed: the p-type side is connected to the negative terminal and the n-type side to the positive terminal. The applied reverse voltage increases the effective barrier potential to $V_{bi} + V_R$, widening the depletion region. Majority carriers are pushed away from the junction, preventing conduction. However, a very small current, called the **reverse saturation current** (I_s), still flows. This current is due to minority carriers (thermally generated electrons in the p-region and holes in the n-region) that are swept across the junction by the electric field. Although I_s is extremely small (in the nanoampere to microampere range), it plays a crucial role in determining the diode's behavior at higher voltages and temperatures.

1.4.3 I-V Characteristics and the Shockley Diode Equation

The current–voltage relationship of a diode can be described by the **Shockley diode equation**:

$$I = I_s \left(e^{\frac{qV}{\eta kT}} - 1 \right), \tag{1.1}$$

where I is the diode current, V is the applied voltage, I_s is the reverse saturation current, q is the charge of an electron (1.602×10^{-19} C), k is Boltzmann's constant (1.38×10^{-23} J/K), T is the absolute temperature in Kelvin, and η is the ideality factor (close to 1 for germanium and around 2 for silicon). This equation shows that in forward bias, the current increases exponentially with voltage, while in reverse bias, the current remains close to $-I_s$ until breakdown occurs.

If the reverse bias is increased beyond a certain critical voltage known as the **breakdown voltage** (V_{br}), the diode enters the breakdown region. Two mechanisms are responsible for breakdown. In **Zener breakdown**, which occurs in heavily doped diodes at low voltages (below about 5–6 V), the electric field across the very thin depletion region is strong enough to pull electrons directly from their bonds by quantum tunneling. In **avalanche breakdown**, which occurs in lightly doped diodes at higher voltages, minority carriers accelerated by the strong electric field collide with atoms in the lattice, knocking loose additional electrons in a process called **impact ionization**. These new carriers are also accelerated, creating a chain reaction and a sudden increase in current. Breakdown itself is not destructive provided the current is limited by an external resistor or circuit element.

1.5 Zener Diode

A **Zener diode** is a specially designed p-n junction diode that is intended to operate safely and reliably in the reverse breakdown region. Unlike a regular diode, where breakdown is normally considered harmful, the Zener diode makes controlled use of this effect.

The most important feature of a Zener diode is its **heavy doping**, which produces a very thin depletion region. Because of this, a relatively small reverse voltage (typically less than 6 V) can create an electric field strong enough to cause Zener breakdown. Above this voltage, known as the **Zener voltage** (V_Z), the diode conducts large amounts of current while maintaining nearly constant voltage across its terminals.

For Zener diodes rated above about 6 V, the dominant breakdown mechanism is avalanche breakdown, but these are still commonly referred to as Zener diodes.

The key characteristic of a Zener diode is therefore its ability to hold the voltage nearly constant once breakdown begins. This makes it especially useful in circuits where a stable reference voltage is required. In practice, a Zener diode is connected in reverse bias across a load, and once the input voltage exceeds V_Z, the diode clamps the voltage at this fixed level, even if the input or load conditions vary.

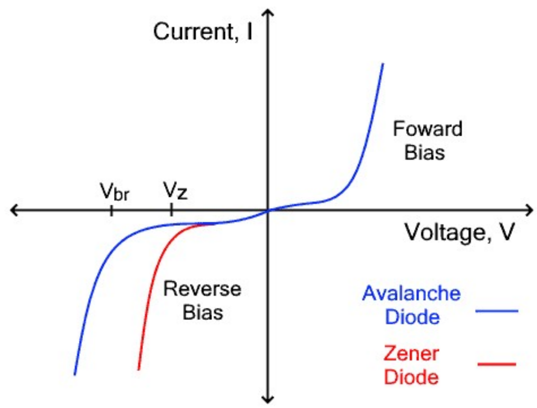

Figure 1.4: VI-Characteristics of Zenor and Avalanche Diodes

Table 1.1: Comparison of Zener and Avalanche Breakdown

Feature	**Zener Breakdown**	**Avalanche Breakdown**
Doping	Heavily doped junction	Lightly doped junction
Depletion Region	Very thin	Relatively wide
Voltage Range	Occurs at low voltages ($<$ 5–6 V)	Occurs at higher voltages ($>$ 6 V)
Mechanism	Quantum mechanical tunneling of electrons across the junction	Impact ionization (carriers collide with atoms and release more carriers)
Current Rise	Sharp and controlled	Sudden but more gradual compared to Zener
Application	Used in Zener diodes for voltage regulation	Common in high-voltage diodes, power devices

1.5.1 Applications

Zener diodes are widely used in modern electronics. Their primary role is in **voltage regulation**, where they act as simple and effective voltage stabilizers. For example, a 5.1 V Zener diode can be used to keep the voltage across a load fixed at that value, despite fluctuations in the supply. They are also employed in **surge protection circuits**, where they clamp sudden overvoltages and protect sensitive electronic components. Beyond this, Zener diodes are used in waveform shaping circuits, voltage limiters, and as reference elements in analog-to-digital converters and other precision applications.

1.6 Transistor (BJT - Bipolar Junction Transistor)

A **Bipolar Junction Transistor (BJT)** is a three–terminal semiconductor device that has become one of the most important elements of modern electronics. Invented in 1947 at Bell Labs, it was the first device that could both amplify electrical signals and act as an electronic switch, paving the way for integrated circuits and the digital revolution. The name "bipolar" comes from the fact that both types of charge carriers are involved in its operation: electrons, which are negatively charged, and holes, which act as positive carriers. Every transistor has three terminals: the emitter, the base, and the collector. These regions are formed by carefully doping a semiconductor such as silicon. Depending on the order in which the p–type and n–type regions are arranged, a transistor may be classified as either an NPN or a PNP type. In practice, NPN transistors are used more widely because electrons, which are their majority carriers, have much higher mobility than holes, giving them better efficiency and high–frequency performance.

Structure of NPN and PNP Transistors

The BJT is built from three semiconductor regions arranged in sequence. In an NPN transistor, a thin and lightly doped p–type base is sandwiched between a heavily doped n–type emitter and a moderately doped n–type collector. Conversely, in a PNP transistor, a thin n–type base lies between two p–type regions. Each of the three regions is designed for a specific function. The emitter is heavily doped to provide a large number of carriers—electrons in the NPN and holes in the PNP. The base is deliberately made very thin and only lightly doped, so that most of the carriers injected from the emitter will diffuse across without recombining. The collector is moderately doped and physically larger than the emitter. Its task is to gather the carriers that make it through the base, and its larger size allows it to dissipate heat and handle higher voltages.

Although NPN and PNP transistors work on the same principle, their polarities and carrier types are opposite. In an NPN transistor, electrons are the majority carriers, and to turn the transistor on the base must be made positive relative to the emitter.

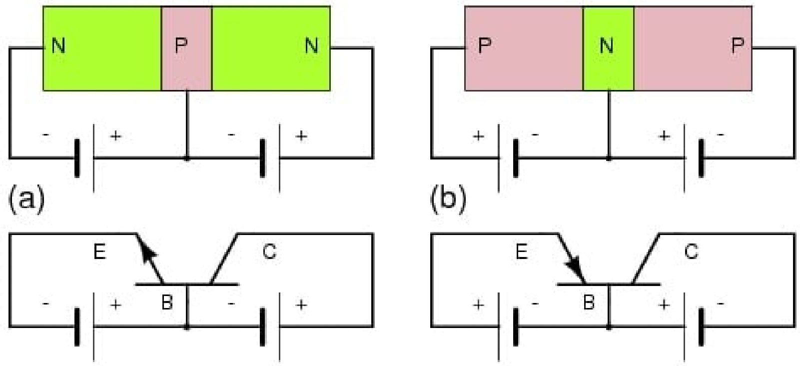

Figure 1.5: Enter Caption

Conventional current then flows from the collector to the emitter. In a PNP transistor, holes are the majority carriers, and the base must be made negative relative to the emitter to switch it on. Here the conventional current flows from the emitter into the collector. Because electrons have higher mobility than holes, NPN transistors generally operate faster and more efficiently, which is why they are more commonly used. A useful way to remember the symbols is that in an NPN transistor the emitter arrow points *out*, while in a PNP transistor the arrow points *in*.

Table 1.2: Comparison of NPN and PNP Transistors

Feature	**NPN Transistor**	**PNP Transistor**
Majority Carriers	Electrons	Holes
Base bias (to turn ON)	Positive with respect to emitter	Negative with respect to emitter
Conventional Current Direction	Collector → Emitter	Emitter → Collector
Symbol	Arrow points **out**	Arrow points **in**
Performance	Higher speed and efficiency (due to high electron mobility)	Lower speed; often used in complementary pairs with NPN
Common Use	More widely used in circuits	Less common, but essential for symmetry in push–pull designs

Working Principle

The essential principle of a transistor is that a very small current flowing into the base terminal is able to control a much larger current flowing between the collector and the emitter. In the case of an NPN transistor, the base–emitter junction is forward biased (requiring about 0.7 V in silicon), while the base–collector junction is reverse biased. The forward bias reduces the barrier at the base–emitter junction, allowing the emitter to inject electrons into the base. Because the base is thin and lightly doped, only a small fraction of these electrons recombine with holes in the base, giving rise to the base current I_B. The majority of electrons pass into the collector, where the reverse–biased base–collector junction sweeps them across, forming the collector current I_C. For a PNP transistor, the process is similar but with opposite polarities: holes are injected from the emitter into the base, and most are collected by the collector.

The three terminal currents are related by the equation

$$I_E = I_B + I_C.$$

Since the base current is very small, the collector current is almost equal to the emitter current. The ability of the transistor to amplify is described by its current gain. In common–base configuration the current gain is

$$\alpha = \frac{I_C}{I_E},$$

which is always slightly less than one. In common–emitter configuration the current gain is

$$\beta = \frac{I_C}{I_B},$$

which typically lies between 20 and 300. These two parameters are related by

$$\beta = \frac{\alpha}{1-\alpha},$$

showing that when α is close to unity, β becomes very large. This explains how a small base current can control a much larger collector current.

1.7 Deriving α and β

Current relations in a BJT (CE view)

In any BJT, the terminal currents satisfy

$$I_E \;=\; I_B \;+\; I_C. \tag{1.2}$$

In the forward-active (amplifier) region, a small base current I_B controls a much larger collector current I_C, while the emitter current I_E is the sum of the two.

Definitions (dc gains)

The two standard dc current gains are defined by

$$\alpha \equiv \left(\frac{I_C}{I_E}\right)_{V_{CB}} \qquad \text{and} \qquad \beta \equiv \left(\frac{I_C}{I_B}\right)_{V_{CE}}, \tag{1.3}$$

where the subscripts indicate the junction voltage held constant during the ratio (for dc, this simply reminds us of the bias condition). Physically, α is the fraction of the emitter current that reaches the collector; β is the multiplication factor relating the collector current to the base drive.

Ideal forward-active case (no leakage)

Start from the current sum (1.2) and the definition of α:

$$\alpha = \frac{I_C}{I_E} \Rightarrow I_C = \alpha I_E.$$

Substitute this I_C into (1.2) to eliminate I_C:

$$I_E = I_B + \alpha I_E \Rightarrow I_B = I_E - \alpha I_E = (1-\alpha)I_E.$$

Solve for I_E and I_C in terms of I_B:

$$I_E = \frac{I_B}{1-\alpha}, \qquad I_C = \alpha I_E = \frac{\alpha}{1-\alpha} I_B.$$

By definition of β,

$$\beta = \frac{I_C}{I_B} = \frac{\alpha}{1-\alpha}. \tag{1.4}$$

Rearranging gives the inverse relation

$$\alpha = \frac{\beta}{1+\beta}. \tag{1.5}$$

These two equations (1.4)–(1.5) establish the fundamental link between the common-base and common-emitter dc gains.

1.7.1 Input and Output Characteristics

The behavior of a transistor can be studied using its input and output characteristics. The input characteristics describe the variation of base current with base–emitter voltage at a constant collector–emitter voltage. This curve resembles that of a diode, since the base–emitter junction is just a forward–biased p–n junction. The output characteristics show how the collector current changes with collector–emitter voltage for various fixed values of base current. These curves reveal three important regions. In the cutoff region, where $I_B = 0$, the collector current is essentially zero. In the active region, the collector current depends mainly on I_B and only weakly on V_{CE}. In the saturation region, when V_{CE} is very small, the transistor no longer behaves like an amplifier but instead like a switch that is fully on.

Modes of Operation

A BJT can function in several distinct modes depending on the biasing of its two junctions. In the **active mode**, the base–emitter junction is forward biased while the base–collector junction is reverse biased, and under this condition the transistor operates as an amplifier because the collector current is controlled by the base current. When both junctions are forward biased, the device enters the **saturation mode**, behaving like a closed switch with the collector–emitter voltage dropping to a very small value ($V_{CE(sat)} \approx 0.2\,\text{V}$). If both junctions are reverse biased, the transistor is in the **cutoff mode**, essentially an open switch with negligible current flow ($I_C \approx 0$). A fourth mode, called the **reverse–active mode**, occurs when the base–emitter junction is reverse biased and the base–collector junction is forward biased; in this case the emitter and collector exchange roles, but the mode is rarely used in practice. A concise summary of these operating conditions is given in Table 1.3.

Table 1.3: Operating modes of a BJT and their characteristics

Mode	**BE Junction**	**BC Junction**	**Typical Behavior / Use**
Active	Forward	Reverse	Amplifier: $I_C \propto I_B$. Used in analog circuits.
Saturation	Forward	Forward	Closed switch: transistor fully ON, V_{CE} very small ($\approx$ 0.2 V).
Cutoff	Reverse	Reverse	Open switch: transistor OFF, $I_C \approx 0$.
Reverse–Active	Reverse	Forward	Transistor works in reverse; rarely used in practice.

1.8 Applications

BJTs are widely used in both analog and digital circuits. In the active mode they act as amplifiers, forming the heart of audio amplifiers, radios, and sensors, where weak signals must be strengthened. In the saturation and cutoff modes they function as switches, forming the basis of digital logic gates, LED drivers, and motor controllers. They are also used in oscillators, waveform generators, and as the building blocks of integrated circuits. Thus, even in the age of MOSFETs and other devices, BJTs remain a fundamental component of electronics.

1.8.1 Practical Applications and Modern Relevance

Analog Applications

- **Audio Amplifiers**: High-fidelity amplification stages
- **Radio Frequency Circuits**: Oscillators and RF amplifiers
- **Sensor Interfaces**: Precision analog signal conditioning
- **Voltage References**: Bandgap reference circuits

Digital and Switching Applications

- **Logic Gates**: Traditional TTL and ECL logic families
- **Power Switching**: Motor drivers, relay controllers
- **LED Drivers**: Constant current sources

Mixed-Signal Applications

- **BiCMOS Technology**: Combines BJT analog performance with CMOS digital density
- **Analog-to-Digital Converters**: Precision conversion circuits
- **Power Management**: Voltage regulators and power controllers

1.8.2 Advantages and Limitations

Key Advantages

- High current drive capability per unit area
- Excellent analog performance with high gain-bandwidth product
- Well-characterized behavior with predictable temperature coefficients
- Superior noise performance compared to MOSFETs in some applications

Limitations

- Higher power consumption due to base current requirement
- Lower input impedance compared to MOSFETs
- More complex biasing requirements
- Limited scalability in very small geometries

The BJT's enduring relevance stems from its unique combination of high gain, excellent linearity, and well-understood physics, ensuring its continued use in precision analog applications despite competition from newer technologies.

1.9 Optoelectronic Devices

Optoelectronic devices are semiconductor components that exploit the interaction between light and electricity. They either convert electrical energy into light or convert incident light into electrical signals. Such devices are central to modern technology, appearing in applications as diverse as communication, imaging, energy generation, and everyday consumer products.

1.9.1 Photodiode

A **photodiode** is a p–n junction diode specially designed to detect light. It is normally operated under reverse bias so that the depletion region is wide. When photons with energy greater than the semiconductor's band gap strike this region, they excite electrons from the valence band to the conduction band, creating electron–hole pairs. The built-in electric field of the depletion region immediately sweeps the electrons toward the n-side and the holes toward the p-side, generating a current known as the **photocurrent**. The magnitude of this current is directly proportional to the intensity of the incident light. Because of this property, photodiodes are widely used as light sensors. Common applications include optical fiber communication systems, where they convert light signals into electrical signals, as well as medical imaging equipment and various light-sensing circuits.

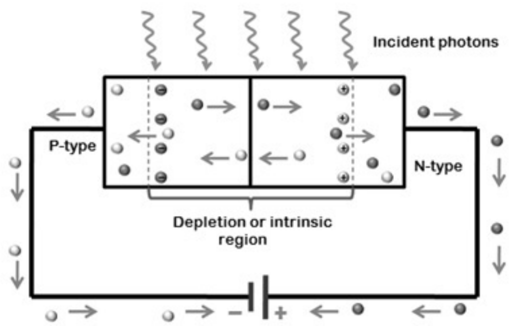

Figure 1.6: Photodiode

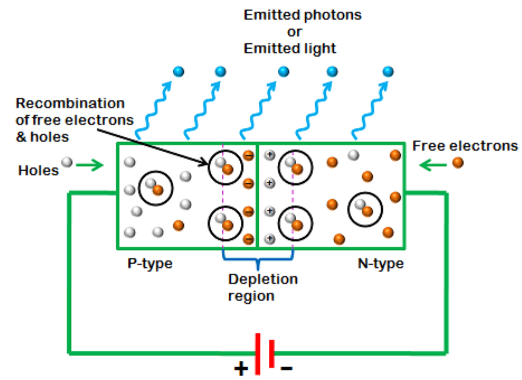

Figure 1.7: LED

1.9.2 Light Emitting Diode (LED)

A **Light Emitting Diode (LED)** is a p–n junction that produces light when forward biased. When a suitable forward voltage is applied, electrons from the n-side and holes from the p-side are injected into the depletion region. In semiconductors with a direct band gap, such as gallium arsenide (GaAs) or gallium nitride (GaN), the recombination of these carriers releases energy in the form of photons instead of heat. The wavelength (and thus the color) of the emitted light depends on the band gap energy of the semiconductor material. For example, GaAs emits infrared light, while GaN can produce blue and green light. LEDs are highly efficient, compact, and durable, which has made them ubiquitous in modern life. Their applications range from simple indicator lamps and seven-segment displays to general-purpose lighting, flat-panel displays, and traffic signals.

1.9.3 Solar Cell

A **solar cell**, also known as a photovoltaic cell, is essentially a large-area p–n junction designed to convert sunlight directly into electricity through the **photovoltaic effect**. When sunlight strikes the cell, photons with sufficient energy excite electrons across the band gap, creating electron–hole pairs. The built-in electric field of the junction separates these carriers, causing electrons to flow through the external circuit while holes move in the opposite direction inside the device. The result is a continuous

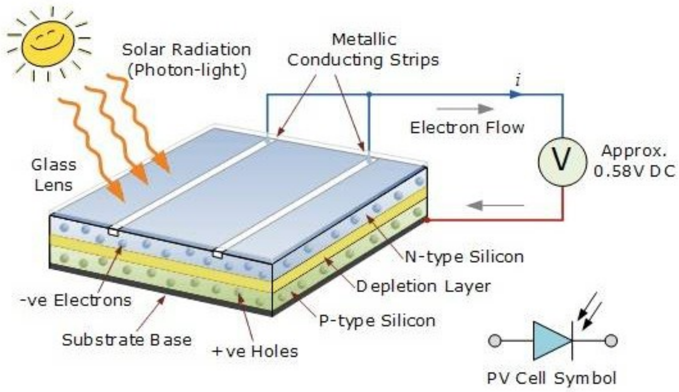

Figure 1.8: Photo voltaic cell construction

Table 1.4: Difference between Photodiode and Solar Cell

Feature	**Photodiode**	**Solar Cell**
Primary Purpose	Light detection (sensor)	Power generation (energy source)
Biasing	Operated under reverse bias	Operated at zero bias (photovoltaic mode) or forward bias
Output	Small photocurrent proportional to light intensity	Usable current and voltage (electrical power)
Optimization	High sensitivity and fast response	High efficiency and maximum energy conversion
Applications	Optical communication, light sensors, medical imaging	Power supply for satellites, homes, solar farms

current and voltage that can be harnessed for useful power. The performance of a solar cell depends on both the band gap of the semiconductor material and the intensity of incident light. Solar cells are used extensively for power generation—from small consumer electronics like calculators and garden lights to large-scale installations powering satellites, homes, and even entire solar farms.